Session 15 (AIML) – Assignment Question

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- College Name: Zeal Polytechnic
- Branch Name: AIML

Experiment: Countries Dataset – Data Reading, Loading, Exploration and Understanding.

OUTPUT:

Step 1: Import Libraries & Upload Dataset. (Google Colab)

```
import numpy as np
import pandas as pd
```

Double-Click (Or enter) to edit

```
[1]
from google.colab import files
uploaded = files.upload()
```

```
Choose files Countries.csv
Countries.csv(text/csv) - 95410 bytes, last modified: 04/07/2026 - 100% done
Saving Countries.csv to Countries.csv
```

Step 2: Read Dataset & Display First 5 Rows

```python
df = pd.read_csv("Countries.csv")

df.head()
```

	country	country_long	currency	capital_city	region	continent	demonym	latitude	longitude	agricultural_land	...	population	women_parliament_seats_pct	rural_population	urban_population	press	democracy_score
0	Afghanistan	Islamic State of Afghanistan	Afghan afghani	Kabul	Southern Asia	Asia	Afghan	33.0	65.0	383560.0	...	41128771	27.01610	30181937	10946834	2.14	2.97
1	Albania	Republic of Albania	Albanian lek	Tirana	Southern Europe	Europe	Albanian	41.0	20.0	11655.5	...	2775634	35.71430	1004807	1770827	2.62	5.98
2	Algeria	People's Democratic Republic of Algeria	Algerian dinar	Algiers	Northern Africa	Africa	Algerian	28.0	3.0	413588.0	...	44903225	8.10811	11328186	33575039	1.71	3.50
3	Andorra	Principality of Andorra	Euro	Andorra la Vella	Southern Europe	Europe	Andorran	42.5	1.5	187.2	...	79824	46.42860	9730	70094	3.17	0.00
4	Angola	People's Republic of Angola	Angolan kwanza	Luanda	Middle Africa	Africa	Angolan	-12.5	18.5	569525.0	...	35588987	33.63640	11359649	24229338	2.24	3.62

5 rows × 64 columns

urban_population	press	democracy_score	democracy_type	median_age	political_leader	title
10946834	2.14	2.97	Authoritarian	12.9	Ashraf Ghani	President
1770827	2.62	5.98	Hybrid regime	33.7	Edi Rama	Prime Minister
33575039	1.71	3.50	Authoritarian	24.0	Abdelmadjid Tebboune	President
70094	3.17	0.00	Unknown	38.9	Xavier Espot Zamora	Head of Government
24229338	2.24	3.62	Authoritarian	12.4	João Lourenço	President

Step 3: Display Last 5 Rows

```
df = pd.read_csv("Countries.csv")
df.tail()
```

	country	country_long	currency	capital_city	region	continent	demonym	latitude	longitude	agricultural_land	...	population
189	Vietnam	Socialist Republic of Vietnam	Vietnamese dong	Hanoi	South-Eastern Asia	Asia	Vietnamese	16.166667	107.833333	123600.0	...	98186856
190	West Bank and Gaza	West Bank and Gaza	Israeli new shekel	Ramallah	Western Asia	Asia	Palestinian	31.900000	35.200000	4449.0	...	5043612
191	Yemen	Republic of Yemen	Yemeni rial	Sana'a	Western Asia	Asia	Yemeni	15.000000	48.000000	234520.0	...	33696614
192	Zambia	Republic of Zambia	New Zambian kwacha	Lusaka	Eastern Africa	Africa	Zambian	-15.000000	30.000000	238360.0	...	20017675
193	Zimbabwe	Republic of Zimbabwe	U.S. dollar	Harare	Eastern Africa	Africa	Zimbabwean	-20.000000	30.000000	162000.0	...	16320537

5 rows × 64 columns

women_parliament_seats_pct	rural_population	urban_population	press	democracy_score	democracy_type	median_age	political_leader	title
30.2605	60123739	38063117	1.04	3.08	Authoritarian	28.4	Nguyễn Phú Trọng	General Secretary of the Communist Party
NaN	1145354	3898258	1.75	4.39	Hybrid regime	15.4	NaN	NaN
0.0000	20491585	13205029	1.27	1.95	Authoritarian	14.8	Abdrabbuh Mansur Hadi	President
15.0602	10857387	9160288	2.15	5.61	Hybrid regime	13.2	Edgar Lungu	President
30.5660	11033499	5287038	1.85	3.16	Authoritarian	14.2	Emmerson Mnangagwa	President

Step 4: Shape of Dataset

```
df = pd.read_csv("Countries.csv")
df.shape
```

```
(194, 64)
```

Step 5: Display Column Names

```
df = pd.read_csv("Countries.csv")
df.columns
```

```
Index(['country', 'country_long', 'currency', 'capital_city', 'region',
       'continent', 'demonym', 'latitude', 'longitude', 'agricultural_land',
       'forest_area', 'land_area', 'rural_land', 'urban_land',
       'central_government_debt_pct_gdp', 'expense_pct_gdp', 'gdp',
       'inflation', 'self_employed_pct', 'tax_revenue_pct_gdp',
       'unemployment_pct', 'vulnerable_employment_pct',
       'electricity_access_pct', 'alternative_nuclear_energy_pct',
       'electricty_production_coal_pct',
       'electricty_production_hydroelectric_pct',
       'electricty_production_gas_pct', 'electricty_production_nuclear_pct',
       'electricty_production_oil_pct', 'electricty_production_renewable_pct',
       'energy_imports_pct', 'fossil_energy_consumption_pct',
       'renewable_energy_consumption_pct', 'co2_emissions',
       'methane_emissions', 'nitrous_oxide_emissions',
       'greenhouse_other_emissions', 'urban_population_under_5m',
       'health_expenditure_pct_gdp', 'health_expenditure_capita',
       'hospital_beds', 'hiv_incidence', 'suicide_rate', 'armed_forces',
       'internally_displaced_persons', 'military_expenditure_pct_gdp',
       'birth_rate', 'death_rate', 'fertility_rate', 'internet_pct',
       'life_expectancy', 'net_migration', 'population_female',
       'population_male', 'population', 'women_parliament_seats_pct',
       'rural_population', 'urban_population', 'press', 'democracy_score',
       'democracy_type', 'median_age', 'political_leader', 'title'],
      dtype='object')
```

Step 6: Dataset Information

```python
df = pd.read_csv("Countries.csv")
df.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 194 entries, 0 to 193
Data columns (total 64 columns):
 #   Column                               Non-Null Count  Dtype
---  ------                               --------------  -----
 0   country                              194 non-null    object
 1   country_long                         194 non-null    object
 2   currency                             194 non-null    object
 3   capital_city                         194 non-null    object
 4   region                               194 non-null    object
 5   continent                            194 non-null    object
 6   demonym                              194 non-null    object
 7   latitude                             194 non-null    float64
 8   longitude                            194 non-null    float64
 9   agricultural_land                    193 non-null    float64
 10  forest_area                          194 non-null    float64
 11  land_area                            194 non-null    float64
 12  rural_land                           194 non-null    float64
 13  urban_land                           194 non-null    float64
 14  central_government_debt_pct_gdp      120 non-null    float64
 15  expense_pct_gdp                      156 non-null    float64
 16  gdp                                  193 non-null    float64
 17  inflation                            184 non-null    float64
 18  self_employed_pct                    179 non-null    float64
 19  tax_revenue_pct_gdp                  159 non-null    float64
 20  unemployment_pct                     179 non-null    float64
 21  vulnerable_employment_pct            179 non-null    float64
 22  electricity_access_pct               194 non-null    float64
 23  alternative_nuclear_energy_pct       138 non-null    float64
 24  electricty_production_coal_pct       138 non-null    float64
 25  electricty_production_hydroelectric_pct 138 non-null    float64
 26  electricty_production_gas_pct        138 non-null    float64
 27  electricty_production_nuclear_pct    138 non-null    float64
 28  electricty_production_oil_pct        138 non-null    float64
 29  electricty_production_renewable_pct  138 non-null    float64
 30  energy_imports_pct                   138 non-null    float64
 31  fossil_energy_consumption_pct        168 non-null    float64
 32  renewable_energy_consumption_pct     192 non-null    float64
 33  co2 emissions                        191 non-null    float64
```

```
 31  fossil_energy_consumption_pct        168 non-null    float64
 32  renewable_energy_consumption_pct     192 non-null    float64
 33  co2_emissions                        191 non-null    float64
 34  methane_emissions                    191 non-null    float64
 35  nitrous_oxide_emissions              191 non-null    float64
 36  greenhouse_other_emissions           184 non-null    float64
 37  urban_population_under_5m            194 non-null    float64
 38  health_expenditure_pct_gdp           190 non-null    float64
 39  health_expenditure_capita            190 non-null    float64
 40  hospital_beds                        192 non-null    float64
 41  hiv_incidence                        133 non-null    float64
 42  suicide_rate                         183 non-null    float64
 43  armed_forces                         174 non-null    float64
 44  internally_displaced_persons         73 non-null     float64
 45  military_expenditure_pct_gdp         163 non-null    float64
 46  birth_rate                           194 non-null    float64
 47  death_rate                           194 non-null    float64
 48  fertility_rate                       193 non-null    float64
 49  internet_pct                         194 non-null    float64
 50  life_expectancy                      191 non-null    float64
 51  net_migration                        194 non-null    int64
 52  population_female                    194 non-null    int64
 53  population_male                      194 non-null    int64
 54  population                           194 non-null    int64
 55  women_parliament_seats_pct           193 non-null    float64
 56  rural_population                     194 non-null    int64
 57  urban_population                     194 non-null    int64
 58  press                                194 non-null    float64
 59  democracy_score                      194 non-null    float64
 60  democracy_type                       194 non-null    object
 61  median_age                           194 non-null    float64
 62  political_leader                     187 non-null    object
 63  title                                187 non-null    object
dtypes: float64(48), int64(6), object(10)
memory usage: 97.1+ KB
```

Step 7: Statistical Summary

```
df = pd.read_csv("Countries.csv")
df.describe()
```

	latitude	longitude	agricultural_land	forest_area	land_area	rural_land	urban_land	central_government_debt_pct_gdp	e
count	194.000000	194.000000	1.930000e+02	1.940000e+02	1.940000e+02	1.940000e+02	194.000000	120.000000	
mean	18.975601	22.027491	2.454551e+05	2.086784e+05	6.675087e+05	6.563711e+05	9777.116531	66.759366	
std	23.876225	66.396389	6.356268e+05	7.824926e+05	1.837107e+06	1.811169e+06	42301.458421	71.806247	
min	-41.000000	-175.000000	4.000000e+00	0.000000e+00	2.027000e+00	3.495450e-02	0.000000	0.000000	
25%	4.000000	-5.000000	6.464000e+03	3.331775e+03	2.355250e+04	2.186562e+04	359.618250	31.951300	
50%	16.583333	21.500000	3.872780e+04	2.528925e+04	1.203750e+05	1.159945e+05	1645.170000	55.426850	
75%	40.000000	50.162500	2.150000e+05	1.236735e+05	5.237000e+05	4.911505e+05	4054.022500	79.539300	
max	65.000000	178.000000	5.285080e+06	8.153120e+06	1.637690e+07	1.622420e+07	522345.000000	687.994000	

8 rows × 54 columns

central_government_debt_pct_gdp	expense_pct_gdp	gdp	...	net_migration	population_female	population_male	population	women_parliament_seats_pct	rural_population	urban_population	press	democracy_score	median_age
120.000000	156.000000	1.930000e+02	...	194.000000	1.940000e+02	1.940000e+02	1.940000e+02	193.000000	1.940000e+02	1.940000e+02	194.000000	194.000000	194.000000
66.759366	30.051403	5.144851e+11	...	-51.407216	2.028316e+07	2.050572e+07	4.078888e+07	25.022994	1.763322e+07	2.315566e+07	2.539330	4.644536	25.661856
71.806247	26.740880	2.307148e+12	...	94525.968598	7.258941e+07	7.607158e+07	1.486470e+08	12.671044	7.664187e+07	7.940393e+07	1.800128	2.818297	9.415569
0.000000	0.000267	6.034940e+07	...	-525116.000000	5.513000e+03	5.799000e+03	1.131200e+04	0.000000	0.000000e+00	5.717000e+03	0.000000	0.000000	10.500000
31.951300	18.371875	1.181390e+10	...	-12242.250000	1.036218e+06	1.044902e+06	2.106358e+06	15.384600	5.896640e+05	1.222244e+06	1.525000	2.722500	16.950000
55.426850	27.337750	4.115390e+10	...	-970.000000	4.502713e+06	4.450049e+06	9.125614e+06	25.252500	2.512382e+06	4.508837e+06	2.400000	5.050000	24.950000
79.539300	35.083500	2.519450e+11	...	2904.250000	1.526606e+07	1.478872e+07	3.031361e+07	33.636400	1.133354e+07	1.621355e+07	2.925000	6.967500	34.050000
687.994000	310.443000	2.546270e+13	...	561580.000000	6.915285e+08	7.311805e+08	1.417173e+09	61.250000	9.088048e+08	8.975784e+08	10.000000	9.870000	50.500000

Step 8: Data Types

```
df = pd.read_csv("Countries.csv")
df.dtypes
```

...

	0
country	object
country_long	object
currency	object
capital_city	object
region	object
...	...
democracy_score	float64
democracy_type	object
median_age	float64
political_leader	object
title	object

64 rows × 1 columns

dtype: object

Step 9: Missing Values

```
df = pd.read_csv("Countries.csv")
df.isnull().sum()
```

...

	0
country	0
country_long	0
currency	0
capital_city	0
region	0
...	...
democracy_score	0
democracy_type	0
median_age	0
political_leader	7
title	7

64 rows × 1 columns

dtype: int64

Step 10: Unique Values

```
df = pd.read_csv("Countries.csv")
df.nunique()
```

	0
country	194
country_long	194
currency	142
capital_city	194
region	22
...	...
democracy_score	146
democracy_type	5
median_age	143
political_leader	187
title	17

64 rows × 1 columns

dtype: int64

Step 11: Random Sample

```
df = pd.read_csv("Countries.csv")
df.sample(5)
```

	country	country_long	currency	capital_city	region	continent	demonym	latitude	longitude	agricultural_land	...	population
89	Kuwait	State of Kuwait	Kuwaiti dinar	Kuwait City	Western Asia	Asia	Kuwaiti	29.500000	45.750000	1500.0	...	4268873
152	Slovenia	Republic of Slovenia	Euro	Ljubljana	Southern Europe	Europe	Slovene	46.116667	14.816667	6104.9	...	2108732
12	Bangladesh	People's Republic of Bangladesh	Bangladeshi taka	Dhaka	Southern Asia	Asia	Bangladeshi	24.000000	90.000000	99010.0	...	171186372
134	Peru	Republic of Peru	Peruvian new sol	Lima	South America	Americas	Peruvian	-10.000000	-76.000000	244783.0	...	34049588
105	Malta	Republic of Malta	Euro	Valletta	Southern Europe	Europe	Maltese	35.833333	14.583333	103.8	...	523417

5 rows × 64 columns

women_parliament_seats_pct	rural_population	urban_population	press	democracy_score	democracy_type	median_age	political_leader	title
6.2500	0	4268873	2.31	3.85	Authoritarian	35.1	Sheikh Sabah Al-Ahmad Al-Jaber Al-Sabah	Emir
40.0000	933093	1175639	3.51	7.50	Flawed democracy	39.5	Janez Janša	Prime Minister
20.8571	103206552	67979820	1.54	5.57	Hybrid regime	22.7	Sheikh Hasina	Prime Minister
40.0000	7249838	26799750	2.59	6.60	Flawed democracy	24.4	Martín Vizcarra	President
27.8481	26825	496592	2.63	8.21	Full democracy	35.3	Robert Abela	Prime Minister

Step 12: Value Counts

```
df = pd.read_csv("Countries.csv")
df["region"].value_counts()
```

region	count
Eastern Africa	17
Western Asia	17
Western Africa	16
Southern Europe	15
Caribbean	13
South America	12
South-Eastern Asia	11
Northern Europe	10
Middle Africa	10
Eastern Europe	10
Western Europe	9
Southern Asia	9

Eastern Europe	10
Western Europe	9
Southern Asia	9
Central America	8
Northern Africa	6
Micronesia	5
Southern Africa	5
Eastern Asia	5
Central Asia	5
Melanesia	4
Polynesia	3
Australia and New Zealand	2
Northern America	2

dtype: int64